Computational Paradigms in Cooperative Multi-Agent Reinforcement Learning: Strategic Adaptability, Cognitive Partner Modeling, and State-of-the-Art Benchmarking in Overcooked Environments

Theoretical Frameworks of Cooperative Multi-Agent Reinforcement Learning

Cooperative multi-agent reinforcement learning (MARL) is a critical subfield of artificial intelligence focused on training decentralized autonomous entities to solve shared tasks[1]. Unlike competitive or mixed-motive environments where agents optimize self-interested outcomes, fully cooperative multi-agent systems operate under a unified, common team reward[1]. This paradigm is mathematically formalized as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP)[4]. A Dec-POMDP is formalized by the tuple:

$$\mathcal{M} = \langle N, S, A, P, R, \Omega, O, \gamma \rangle$$

where N is a finite set of n agents, S represents the global environmental state space, and $A = \prod_{i=1}^{n} A_i$ is the joint action space[4]. The transition dynamics are governed by the probability function $P : S \times A \to \Delta(S)$ [4]. Each agent i cannot directly observe the true environmental state $s \in S$; instead, it receives a local, partial observation $o_i \in \Omega_i$ generated by the observation distribution $O : S \times A \to \Delta(\Omega_i)$ [4]. The team receives a joint reward $R : S \times A \to \mathbb{R}$ at each decision step, and the cooperative objective is to discover a joint policy $\pi = (\pi_1, \ldots, \pi_n)$ that maximizes the expected discounted return[6]:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \mathbf{a}_t) \right]$$

During deployment, agents operate under a Centralized Training with Decentralized Execution (CTDE) framework, meaning they must select individual actions $a_{i,t} \in A_i$ based solely on their localized action-observation history $h_{i,t} \in \mathcal{H}_i$ without direct access to the private observations or internal policies of their teammates[1].
Historically, cooperative policies have been computed using Self-Play (SP), where identical copies of an agent co-adapt within a single training loop[8]. While Self-Play is capable of identifying highly optimized joint trajectories, it routinely suffers from "convention overfitting"[10]. When a game possesses multiple symmetric optimal equilibria, the co-adapting agents break these symmetries arbitrarily to converge to a highly specific, idiosyncratic convention[9].
Because these conventions are highly fragile, pairing a Self-Play agent at test time with an independently trained partner typically results in complete coordination failure[10]. This vulnerability has led to the formalization of several robust coordination paradigms, summarized below in a comparative structural framework.

Coordination Paradigm	**Mathematical and Operational Objective**	**Partner Assumption at Test Time**	**Core Evaluation Metric**
Self-Play (SP) [cite: 8, 9]	$\max_\pi J(\pi, \pi)$	Identical copy of the self, trained in the exact same execution silo[9].	Self-Play Joint Return[11].
Zero-Shot Coordination (ZSC) [cite: 6, 13]	$\max_{\pi_{ego}} \mathbb{E}_{\pi_{partner} \sim \Pi_{alg}} [J$	Independently trained runs of the same algorithm with disjoint initial seeds[6].	Intra-Algorithm Cross-Play (XP)[6].
Ad-Hoc Teamwork (AHT) [cite: 15, 16]	$\max_{\pi_{ego}} \mathbb{E}_{\pi_{partner} \sim \Pi_{held-o}}$	Unfamiliar partners, potentially trained via distinct algorithms or representing humans[8].	Inter-Algorithm Cross-Play (Inter-XP)[12].

Few-Shot Coordination (FSC) [cite: 13]	$\min_{\pi_{ego}} \mathrm{Regret}(\pi_{ego},$	Adaptable or static partners over K episodes of online interaction[13].	Adaptation Regret (convergence rate over steps)[13].

In complex settings, zero-shot coordination may be structurally impossible without some mechanism for rapid online adaptation[13]. This has motivated research into Few-Shot Coordination (FSC), where performance is evaluated using "adaptation regret," measuring an agent's ability to minimize coordination loss over a limited series of interactions with a held-out pool of partners[13].

Crucially, empirical evaluations in games like Hanabi reveal a stark trade-off: state-of-the-art ZSC algorithms such as Off-Belief Learning (OBL) can exhibit high zero-shot performance but struggle to adapt online, whereas naive Independent Q-Learning (IQL) agents can adapt to novel partners just as rapidly as OBL, indicating that robust coordination requires balancing static zero-shot alignment with dynamic online adaptation[13].

Strategic Dimensions and Design Variables for Collaborative Game Agents

Designing artificial agents capable of real-time collaboration with novel partners requires addressing several environmental, cognitive, and behavioral variables[19]. These dimensions govern the transition from co-adapted, closed-loop systems to flexible, generalist collaborators[21].

Physical State Coverage versus Strategic Coordination

A critical issue in cooperative MARL is determining whether coordination failures stem from a lack of strategic alignment or simple physical out-of-distribution (OOD) state transitions[11]. In many collaborative games, agents develop highly synchronized movement patterns and precise action timings during Self-Play[11]. When paired with an unfamiliar partner at test time, minor variations in the partner's speed or path planning cause the ego agent to encounter physical configurations it has never seen before[11]. Because standard deep learning policies are highly sensitive to OOD observations, these minor physical perturbations can cause the agent to freeze, oscillate, or fail entirely[11].

Research by Gessler et al. has demonstrated that the self-play to cross-play (SP-XP) gap in many deterministic cooperative environments is primarily a consequence of insufficient physical state coverage during training rather than sophisticated strategic mismatch[5]. By implementing a state-augmentation mechanism—which stores states from agent interactions in a trajectory buffer and uses them as random initial states during self-play—the SP-XP gap can be nearly closed[11]. This indicates that robust collaborative agents must first achieve exhaustive physical coverage of the state space before more complex, cognitive-level strategic alignment can occur[11].

Theory of Mind and Recursive Cognitive Alignment

Theory of Mind (ToM) represents the cognitive capability of an agent to reason about the latent mental states, beliefs, desires, and strategic intentions of other actors[7]. In collaborative settings, explicitly equipping agents with ToM enables them to transition from reactive policies to proactive coordination[7]. This is typically modeled using recursive reasoning hierarchies[7]:

- **0th-Order ToM ($k = 0$)**: The agent treats its partner as a passive, non-reasoning element of the environment, optimizing its actions based purely on direct state observations[25].
- **1st-Order ToM ($k = 1$)**: The agent models the partner as a 0th-order agent, explicitly estimating the partner's goals and predicting their next actions in order to select a complementary response[7].
- **Higher-Order ToM ($k \geq 2$)**: The agent recursively assumes that the partner is also reasoning about the ego agent's mental state (e.g., "Agent A believes that Agent B expects Agent A to yield")[7].

While recursive reasoning can enhance coordination, research shows that misaligned ToM orders—mismatches in the depth of reasoning between the ego agent and its partner—can severely degrade performance[7]. For instance, if a highly sophisticated $k = 2$ agent is paired with a simple, non-adaptive $k = 0$ partner, the $k = 2$ agent may over-interpret the partner's actions, executing erratic "signaling" behaviors that lead to coordination failure[7].

To prevent this, state-of-the-art designs utilize Adaptive Theory of Mind (A-ToM) frameworks, where the ego agent dynamically infers the partner's likely ToM order from historical interaction steps and aligns its own reasoning depth in real time[7].

Additionally, frameworks like ProToM combine Bayesian inverse planning with expectation utility maximization to communicate context-grounded, timely feedback, minimizing communication overhead while maximizing alignment[28].

Information Structures and Observability

The complexity of cooperative interactions is heavily dictated by the observability of the environment[22].

Under symmetric information and full observability, agents can independently calculate joint action values, reducing the coordination problem to a localized matching exercise[10].

However, under partial observability and asymmetric information, agents must infer hidden state variables through their partner's behavior[22]. This requires the design of explicit communication channels or implicit signaling strategies, such as "demoing" (where an informed agent executes specific physical actions to reveal a hidden recipe or target location to an uninformed partner)[11].

Risk Sensitivity under Behavioral Uncertainty

Real-world human collaboration is rarely risk-neutral[30]. Human partners exhibit varying levels of risk aversion or risk seeking based on their subjective utility models[10].
To formalize this in collaborative game agents, Smith and Zhang introduced a modification of cooperative MARL based on Cumulative Prospect Theory (CPT), which mathematical models decision-making under uncertainty and biased risk preferences[30]. By optimizing policies using CPT-weighted reward structures (such as the MARSRL algorithm based on Double Deep Q-Networks), agents can adapt on the fly to risk-averse, risk-seeking, or risk-neutral partners[30]. This capability is critical in layouts where completing a task requires a coordinated, high-risk physical maneuver[30].

Robustness Evaluation and Diagnostic Metrics

To prevent collaborative agents from developing fragile, highly specialized strategies, modern evaluation frameworks incorporate both subjective human feedback and automated diagnostic testing[19]. Knott et al. established that cooperative agents should be systematically evaluated against automated edge-case unit tests that simulate adversarial partner failures or sudden environmental shifts[32].
Furthermore, research utilizing AgentCollabBench and Collab-Overcooked has introduced process-oriented metrics to evaluate collaboration beyond simple game scores[19]. Rather than relying solely on sparse reward signals, these frameworks use the Trajectory Efficiency Score (TES) and the Incremental Trajectory Efficiency Score (ITES) to measure the fluency, task allocation efficiency, and coordination overhead of human-agent and agent-agent interactions[19].

Architectural Paradigms and Algorithmic Solutions

To solve Dec-POMDP coordination challenges, the MARL community has developed several deep learning architectures and training pipelines[2]. These approaches span value function factorization, multi-stage population-based diversity training, and emerging in-context learning models[4].

Value Function Factorization under Centralized Training with Decentralized Execution

The CTDE paradigm is a widely adopted standard for cooperative MARL[4]. During centralized training, algorithms have access to the global environmental state S and all agents' observations to compute a centralized joint action-value function $Q_{tot}(s, \mathbf{a})$ [1]. During deployment, the centralized mixing components are discarded, leaving only decentralized individual policies $\pi_i(a_i|h_i)$ or individual value networks $Q_i(h_i, a_i)$ [1].

Value-Decomposition Networks (VDN)

VDN represents the simplest form of factorization, assuming that the joint action-value function is a linear summation of the individual agent utilities[1]:

$$Q_{tot}(s, \mathbf{a}) = \sum_{i=1}^{n} Q_i(h_i, a_i)$$

This assumption ignores complex, state-dependent agent-agent interactions, limiting VDN's effectiveness in environments with strong inter-agent coupling[1].

QMIX

QMIX relaxes the strict linearity of VDN by enforcing a monotonicity constraint, ensuring that the joint action-value is monotonic with respect to each individual utility[4]:

$$\frac{\partial Q_{tot}(s, \mathbf{a})}{\partial Q_i(h_i, a_i)} \geq 0, \quad \forall i \in \{1, \ldots, n\}$$

This condition guarantees that the decentralized argmax operator distributes perfectly over individual utilities, enabling decentralized execution without the need for online communication[34]:

$$\underset{\mathbf{a}}{\text{argmax}}\ Q_{tot}(s, \mathbf{a}) = \left(\underset{a_1}{\text{argmax}}\ Q_1(h_1, a_1), \ldots, \underset{a_n}{\text{argmax}}\ Q_n(h_n, a_n) \right)$$

QMIX implements this constraint using a centralized mixing network with non-negative weights generated by hypernetworks conditioned on the global state s [34].
However, because monotonic mixing cannot represent joint action-value functions where an agent's optimal choice depends on another's specific exploratory action, QMIX can perform poorly in highly cooperative, non-monotonic environments[34]. This limitation has led to weighted variants, such as WQMIX, CW-QMIX, and OW-QMIX, which attempt to approximate optimal joint action allocations[34].

QPLEX

QPLEX addresses the representation limitations of QMIX by utilizing a dueling network architecture to factorize the joint action-value function[4]. By separating the state-value function from individual advantage functions, QPLEX can represent a broader class of cooperative games without enforcing the monotonic restriction of QMIX[4].

Multi-Stage Population-Based Training Paradigms

Because standard Self-Play often fails to generalize, state-of-the-art zero-shot coordination algorithms utilize multi-stage training pipelines[10]. These methods focus on constructing a highly diverse population of partner policies in the first stage, and then training a robust "ego" policy to act as the optimal best response to that population in the second stage[10].

Fictitious Co-Play (FCP)

Developed by Strouse et al., FCP is a robust, computationally elegant two-stage framework[3].

In the first stage, the algorithm trains N independent Self-Play agent pairs using distinct random seeds for neural network initialization[9]. These independent runs force the pairs to converge to different arbitrary symmetry-breaking conventions[9]. Throughout this training phase, the algorithm saves policy checkpoints[9]. The final checkpoints represent highly skilled "expert" partners, while earlier checkpoints capture different lower-skill strategies[9].

In the second stage, the entire pool of generated partners and their historical checkpoints is frozen[9]. A single "ego" agent is then trained via reinforcement learning to act as the optimal best response to the uniform distribution over this frozen partner pool[9]:

$$\pi_{ego} = \underset{\pi}{\operatorname{argmax}} \, \mathbb{E}_{\pi_p \sim \mathcal{P}_{pool}} \left[J(\pi, \pi_p) \right]$$

Because the partners' parameters are frozen, the ego agent cannot force them to adapt; instead, it must learn to dynamically identify the partner's strategy and skill level from its online actions and adapt its own behavior accordingly[9].

Maximum Entropy Population-Based Training (MEP)

MEP improves upon FCP by explicitly optimizing for diversity during the population generation phase[38]. It defines a Population Diversity (PD) objective that combines the average individual policy entropy with the pairwise Kullback-Leibler (KL) divergence between all policies in the population[40]:

$$PD\left(\{\pi^{(1)}, \ldots, \pi^{(n)}\}, s_t\right) = \frac{1}{n}\sum_{i=1}^{n} H\left(\pi^{(i)}(\cdot|s_t)\right) + \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} D_{KL}\left(\pi^{(i)}(\cdot|s_t) \parallel \pi^{(j)}(\cdot|s_t)\right)$$

To optimize this complex objective efficiently, MEP derives a safe and computationally bounded surrogate lower bound termed the **Population Entropy (PE)**, defined as the entropy of the mean policy of the population[40]:

$$PE\left(\{\pi^{(1)}, \ldots, \pi^{(n)}\}, s_t\right) = H\left(\bar{\pi}(\cdot|s_t)\right), \quad \text{where} \quad \bar{\pi}(a_t|s_t) = \frac{1}{n}\sum_{i=1}^{n} \pi^{(i)}(a_t|s_t)$$

By incorporating the PE as an auxiliary reward bonus during the first-stage co-training phase,

the population is forced to diverge into mutually complementary, highly exploratory strategies[40].
In the second stage, rather than training the robust ego agent against a uniform distribution of these partners, MEP employs a learning-progress-based prioritized sampling mechanism[38]. This prioritized sampling is dynamically adjusted based on the training progress, skewing the sampling distribution to focus on partners where the ego agent is currently underperforming[40]:

$$p(\pi^{(i)}) \propto \exp\left(-\beta \mathbb{E}[J(\pi_{ego}, \pi^{(i)})]\right)$$

This mechanism prevents the ego agent from exploiting easy-to-collaborate partners while neglecting difficult coordination conventions[40].

Hidden-Utility Self-Play (HSP)

HSP is designed to address a critical limitation of both FCP and MEP: the assumption that all potential test-time partners are fundamentally rational and actively optimizing the ground-truth game reward[10]. In reality, human players exhibit substantial cognitive and behavioral biases, often optimizing implicit personal preferences[10].
To capture this non-rational strategy space, HSP explicitly models human biases as hidden reward functions in the self-play objective during the population generation phase[10]. The reward space is approximated as a linear combination of the ground-truth environment reward $R(s, a)$ and a vector of hand-designed, task-relevant event features $\phi(s, a)$ [44]:

$$R_w(s, a) = R(s, a) + \mathbf{w}^T \phi(s, a)$$

The weight vectors $\mathbf{w}$ are sampled from a prior distribution $\hat{P}_R$, and independent self-play agents are trained to optimize these biased reward formulations[10]. By executing a random search over this approximated hidden-utility space, HSP generates an augmented policy pool that covers a wide spectrum of biased, non-adaptive behaviors[10]. Training the ego agent against this biased pool yields exceptional robustness and assistance capabilities when paired with real humans[10].

Cooperative Open-ended LEarning (COLE)

COLE addresses the issue of "cooperative incompatibility in learning"[8]. When training an ego agent against a population, standard algorithms can suffer from learning loss where optimizing for average population performance causes the agent to lose its ability to coordinate with specific, highly specialized strategies[8]. COLE formalizes this by constructing Graphic-Form Games (GFGs)[8].
In a GFG, the population of strategies is represented as nodes in a directed graph, where the directed edge weights W_{ij} represent the cross-play payoff between strategy i and strategy

j [47]. Using graph-theoretic centrality metrics (such as in-degree centrality or Shapley value formulations), COLE mathematically pinpoints the cooperative capability and compatibility profile of each strategy[8].
COLE proposes two practical solvers[8]:

- $COLE_{SV}$: Uses the cooperative game theory concept of the Shapley value to evaluate the marginal contribution of each strategy to the population's overall coordination robustness, determining a precise "cooperative incompatibility distribution"[8].
- $COLE_{R}$: Replaces the computationally expensive Shapley value calculations with a simplified solver based on average node payoffs to estimate cooperative capacity[8].

The trainer then dynamically constructs a mixture distribution of these incompatible partners and trains the ego policy to approximate the best response to this mixture, theoretically guaranteeing convergence to a locally optimal, universally compatible strategy[8].

In-Context Learning and Transformer-Based Policies

To bypass the computational cost of training explicit policy populations, recent research has explored transformer-based architectures that learn to coordinate entirely *in-context*[15].
By framing multi-agent interaction as a sequence modeling problem, algorithms such as Algorithm Distillation (AD) and Decision-Pretrained Transformers (DPT) can achieve rapid test-time adaptation without updating their neural network weights online[15].

- **Algorithm Distillation (AD)**: AD is trained on learning histories generated by source RL algorithms interacting with various teammates[15]. A training sample consists of a sequence of episodes reflecting the source algorithms' learning progress from random to expert[15]. AD minimizes:

$$\mathcal{L}_{\mathrm{AD}}(\theta) = \mathbb{E}_{\tau \sim \mathcal{D}_{\text{learning}}} \left[- \sum_{t} \log \pi_\theta(a_t \mid h_t) \right]$$

  By modeling policy improvements across episodes within its historical context h_t , AD learns to perform in-context meta-exploration, identifying the teammate's strategy and adapting to maximize returns[15].
- **Decision-Pretrained Transformer (DPT)**: DPT is trained on expert trajectories to perform in-context task inference[15]. The objective is to predict the optimal action a_t^* given a context h_t that characterizes the teammate's behavior[15]:

$$\mathcal{L}_{\mathrm{DPT}}(\theta) = \mathbb{E}_{\tau^* \sim \mathcal{D}_{\text{expert}}} \left[- \sum_{t} \log \pi_\theta(a_t^* \mid h_t) \right]$$

Unlike AD, DPT does not perform in-context meta-exploration; instead, it treats the interaction history as a prompt to identify the underlying MDP and directly query the optimal policy, achieving zero-shot adaptation[15].

State-of-the-Art Analysis in the Overcooked Benchmark Ecosystem

The cooperative cooking simulator **Overcooked-AI** has established itself as the primary benchmark for evaluating zero-shot human-AI and agent-agent coordination[11]. Based on the popular video game *Overcooked*, the environment requires two agents to cooperate in a grid-world kitchen to prepare and deliver soups[48]. Preparing a soup requires placing up to three ingredients (e.g., onions or tomatoes) into a pot, waiting for it to cook, and harvesting the cooked soup with a plate to deliver it to the serving station[45]. This simple setup abstracts complex collaborative challenges, such as task division, spatial navigation, and dynamic coordination[48].

Classic Overcooked-AI Layouts

The standard Overcooked-AI benchmark evaluates agents across five classic layouts, each isolating a specific structural coordination challenge[50]:

- **Cramped Room**: A small, obstacle-free kitchen with shape (5, 4)[50]. Because of the highly confined space, agents must learn precise path planning, collision avoidance, and dynamic yielding to prevent physical deadlocks[51].
- **Asymmetric Advantages**: A layout where each agent is confined to a distinct half of the kitchen, but access to resources is asymmetric[51]. This layout tests the agents' ability to recognize and exploit specialized advantages (e.g., one agent has faster access to onions, while the other is closer to the cooking pots)[51].
- **Coordination Ring**: A kitchen arranged in a ring layout[51]. Agents must coordinate their movement directions (e.g., traveling clockwise vs. counter-clockwise) to avoid physical collisions in the narrow corridors while transporting ingredients and dishes[51].
- **Counter Circuit**: A layout where a central counter completely divides the two agents, preventing them from accessing each other's workspaces[31]. To complete soups, agents must pass ingredients, plates, and cooked soups across the counter, requiring tight coordination[31].
- **Forced Coordination**: A layout that enforces strict, sequential role assignment[52]. Because one agent is blocked from accessing the ingredients and the other is blocked from accessing the pots, they cannot complete the task independently, requiring a continuous hand-off sequence[52].

OvercookedV2: Redefining Zero-Shot Coordination

As MARL research progressed, a fundamental limitation in the original Overcooked-AI benchmark was uncovered[11]. Gessler et al. demonstrated that the self-play to cross-play

(SP-XP) gap in the original game was largely an artifact of **poor physical state coverage** during self-play training, rather than a deep strategic coordination challenge[5]. Because the environment is fully observable and completely deterministic, self-play agents learn highly synchronized, open-loop trajectories[11]. Minor physical offsets at test time cause the agents to encounter OOD states, leading to failure[11]. By applying simple state-augmentation during training, the SP-XP gap can be nearly closed across all five classic layouts using standard, non-adaptive algorithms[11].

To address these shortcomings, **OvercookedV2** was developed within the JaxMARL framework, introducing structural elements that cannot be solved by simple state-coverage[5]:

- **Configurable Stochasticity**: Transitions are no longer deterministic, preventing agents from relying on fixed, open-loop schedules and forcing the use of closed-loop, adaptive feedback policies[22].
- **Partial Observability**: Agents are given limited local view fields, forcing them to maintain internal beliefs about their partner's actions and location[22].
- **Asymmetric Information (Grounded Coordination)**: In layouts like *Asymmetric Advantages V2*, only one agent can view the dynamic recipe indicator[22]. This forces the informed agent to actively "demo" the recipe by placing correct ingredients in the pot, while the uninformed agent must stall, observe, and interpret these actions before assisting[22].
- **Test-Time Protocol Formation**: Introduces layouts where it is impossible to coordinate via a pre-trained, static policy[11]. Independently trained agents must dynamically negotiate and form communication or coordination protocols online at test time[11].

Empirical evaluations on OvercookedV2 show that state-of-the-art ZSC algorithms (such as FCP and Other-Play) experience a severe drop in performance[23]. When paired with novel partners, their rigid, pre-trained conventions break down completely, highlighting a major open challenge for adaptive, online-coordinating MARL algorithms[23].

The Overcooked Generalisation Challenge (OGC)

While standard ZSC focuses on adapting to unfamiliar partners in a *fixed* layout, real-world human-AI teaming requires generalization across both **partner policies** and **environmental layouts**[15]. The **Overcooked Generalisation Challenge (OGC)** addresses this by bridging Unsupervised Environment Design (UED) with zero-shot multi-agent cooperation[53].

OGC integrates the benchmark with GPU-accelerated Dual Curriculum Design (DCD) frameworks via the *minimax* suite, allowing the automatic generation of billions of solvable kitchen configurations[38]. The challenge evaluates agents across two distinct generalization tracks[15]:

- **Track 1 (Teammate Generalization)**: Evaluates adaptation to completely held-out, novel partner policies within familiar, trained environmental layouts[15].
- **Track 2 (Layout Generalization / Double Generalization)**: The most demanding track, pairing novel, unfamiliar partner policies with entirely unseen, procedurally generated environmental layouts[15].

Evaluations on the OGC reveal a critical limitation of current SOTA algorithms[54]. Standard DCD generators (like PAIRED, Robust PLR, and ACCEL) struggle in the complex, highly interactive design space of Overcooked[54]. Because the environmental generator must account for agent-agent interaction dynamics, standard generators fail to produce structured, progressive curricula, resulting in uncooperative, brittle policies[54]. Only advanced architectures incorporating Soft Mixture-of-Experts (SoftMoE) modules combined with PAIRED generators show minor generalization capabilities, establishing the OGC as a primary testbed for cooperative AI research[54].

Comparative Synthesis of SOTA Algorithms on Overcooked Benchmarks

To illustrate the strengths, operational mechanisms, and failure modes of current cooperative algorithms, the following table synthesizes performance across the classic Overcooked-AI, OvercookedV2, and OGC benchmarks.

Algorithm	Primary Training Paradigm	Partner Diversity Mechanism	Empirical SOTA Performance on Overcooked Benchmarks	Primary Coordination Failure Modes
Self-Play (SP) [cite: 8, 9]	Standard co-adaptation within a single centralized training loop[9].	None. Relies purely on the co-evolving partner policy[9].	Computes highly optimized, near-optimal joint trajectories in classic layouts, but fails in cross-play[9].	**Convention Overfitting**: Fails immediately when paired with any partner using slightly different movement conventions[10].
Fictitious Co-Play (FCP) [cite: 9]	Best-response training against a frozen population of self-play checkpoints[9].	Diversifies by training independent SP runs with varying random seeds	Highly robust in classic layout cross-play; provides a stable,	**Exploration Limits**: Cannot coordinate when test-time partners deviate

		and checkpointing skill levels[9].	assistive partner for humans[9].	significantly from rational, self-play-like trajectories[10].
Maximum Entropy Population (MEP) [cite: 38, 40]	Best-response training against a population optimized for behavioral diversity[38].	Incorporates a Population Entropy (PE) auxiliary reward bonus to maximize policy KL-divergence [40].	Outperforms FCP in complex, non-monotonic layouts (e.g., Counter Circuit); highly robust in ZSC[38].	**Computational Overhead**: Stage-one training is highly intensive, often exceeding 15 hours on modern GPUs[41].
Hidden-Utility Self-Play (HSP) [cite: 10, 44]	Best-response training against a population with perturbed reward weights[10].	Evaluates policies against augmented, biased reward spaces ($R_w = R +$)[44].	Consistently preferred by human players; achieves 10x higher scores when cooperating with biased partners[10].	**Feature Dependency**: Highly dependent on the design of the event features ϕ to capture relevant human biases[37].
Cooperative Open-ended Learning (COLE) [cite: 8, 47]	Iterative GFG reconstruction and compatibility-weighted optimization[8].	Uses graph-theoretic solvers ($COLE_{SV}$, $COLE_R$) to map the population's compatibility landscape[8].	Exceeds MEP and FCP in complex layouts; effectively overcomes cooperative incompatibility [8].	**Algorithmic Complexity**: High training complexity due to continuous GFG updates and solver steps[8].
Off-Belief Learning (OBL)	Hierarchical belief-decoupled	Decouples past action interpretation	Theoretically converges to a unique	**Covariate Shift**: Extremely

[cite: 58]	reinforcement learning[58].	from future action optimization across levels[58].	coordination policy, but underperforms in ad-hoc teamwork[12].	fragile to test-time distribution shifts when paired with non-OBL or human partners[6].
Behavior Augmentation, Simulation, and Selection (BASS) [cite: 60]	Joint behavioral diversification, simulated planning, and real-time selection[60].	Augments physical behavior and simulates environmental outcomes to enhance partner adaptability[60].	Outperforms prior SOTA under low-level motion control and physical environment layouts[60].	**Low-Level Focus**: Designed primarily for continuous, physics-based settings; less effective in abstract strategic tasks[60].
Decision-Pretrained Transformer (DPT) [cite: 15]	Sequence modeling of expert/learning trajectories[15].	Learns in-context by modeling diverse teammate strategies in its historical context[15].	Achieves rapid, zero-shot test-time adaptation without online weight updates[15].	**Data Dependency**: Requires exhaustive expert trajectory datasets for effective pre-training[15].

Strategic Synthesis and Future Outlook

The evolution of cooperative multi-agent reinforcement learning highlights a clear developmental trajectory[11]. The field is transitioning from closed-loop, specialized coordination (Self-Play and value decomposition) to open-loop, population-based generalization (FCP, MEP, HSP), and finally to open-ended, double-generalization challenges (OvercookedV2, OGC, ad-hoc teamwork)[8].

This progression reveals several key insights:

First, training a static, generalist best-response policy against a frozen training pool is inherently limited[38]. While robust, these static models fail when paired with highly dynamic partners or humans who rapidly adapt and change their strategies online[13]. True ad-hoc

coordination requires agents that can perform real-time, in-context learning[15].
Second, the structural constraints of the training environment dictate the cooperative limits of the policy[11]. In deterministic, fully observable settings, agents learn to exploit environmental dynamics to bypass the need for active coordination[11]. Forcing agents to develop true, robust collaboration requires the integration of asymmetric information, stochastic transitions, and partial observability, as realized in OvercookedV2[11].
Finally, the future of collaborative MARL lies in the intersection of neural policy learning, cognitive partner modeling, and unsupervised environment design[7]. By combining transformer-based in-context learning with dynamic curriculum generation (such as the Overcooked Generalisation Challenge), researchers can train agents that do not merely master specific, localized conventions, but instead learn general cooperative norms that enable seamless collaboration with novel partners across any environment[15].

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